Introduction

The rapid expansion of digital media and online news platforms has transformed the way information is accessed and disseminated worldwide. News is now instantly available through websites, mobile applications, and social networking platforms. However, this accessibility has also facilitated the widespread circulation of fake news, which can mislead the public, influence opinions, and create social unrest. Fake news refers to false or misleading information presented in the format of legitimate news content.

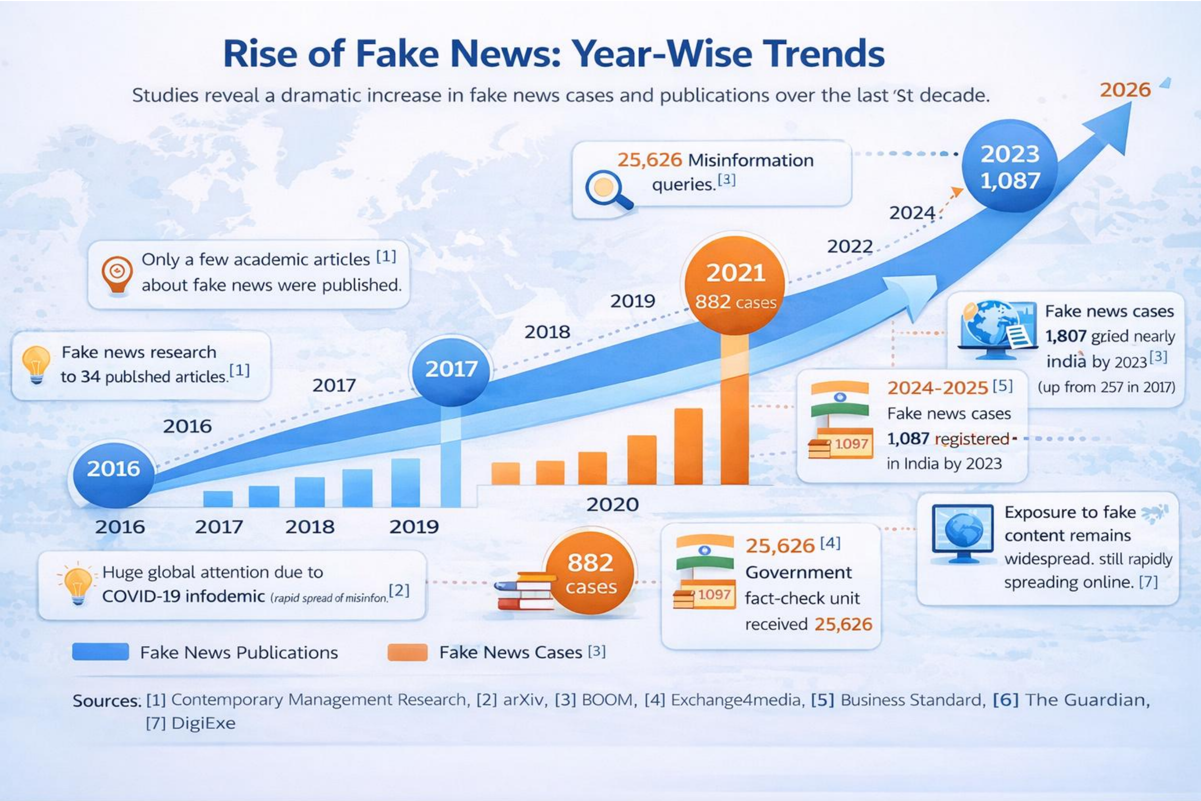
The large volume of digital information and the speed at which it spreads make manual verification ineffective and impractical. Studies indicate that false information propagates more rapidly than verified content on social media platforms, thereby increasing its societal impact. The growing influence of misinformation highlights the need for automated and intelligent systems capable of detecting and limiting the spread of fake news.

To address this challenge, the proposed system presents a web-based solution that integrates news aggregation with fake news detection. The system is not limited to classification alone; it functions as a unified news aggregation and verification portal. News articles are collected from multiple trusted sources, organized into relevant categories such as politics, sports, technology, and entertainment, and displayed within a single interface. This centralized structure eliminates the need for users to visit multiple news platforms, thereby improving accessibility and usability.

The collected news content undergoes preprocessing using Natural Language Processing (NLP) techniques, including tokenization, stopword removal, and text cleaning. The processed text is then converted into numerical sequences and analyzed using a Long Short-Term Memory (LSTM) deep learning model implemented with TensorFlow and Keras. The model evaluates contextual and sequential patterns within the text to determine whether the news is real or fake.

The web application provides a practical and user-friendly solution by enabling users to both browse categorized news articles from multiple sources in one portal and verify authenticity in real time. In addition to automated analysis of scraped content, the system allows users to manually input any news text for verification. This dual functionality enhances convenience, promotes informed decision-making, and contributes to reducing the impact of misinformation in digital environments.

Year Wise Fake News Statistics



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